

Question number	Answer	Mark
7(a)	C they can produce some types of cell	(1)

Question number	Answer	Mark
7(b)	A description that includes the following points: • ribosomes involved in translation resulting in two different polypeptides (1) • endoplasmic reticulum involved in folding each polypeptide into tertiary structure (1) • endoplasmic reticulum involved in transporting each polypeptide (1)	(3)

Question number	Answer	Mark
7(c)(i)	A description that includes the following points: • α polypeptide increases to maximum by 6 months before birth and then stays constant (1) • β polypeptide rises slowly during pregnancy but then increases rapidly after birth (1) • γ polypeptide increases to maximum by 6 months before birth and then drops rapidly after birth (1)	(3)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An explanation that includes the following points: • gene for α polypeptide remains switched on as this polypeptide is present both before and after birth (1) • gene for γ polypeptide is switched off at birth so levels fall (1) • gene for β polypeptide switched on during pregnancy so levels rise (1) • credit given for an epigenetic mechanism that switches {on / off} gene expression (1)	For example, {DNA / histone} methylation switches off {genes / gene for γ polypeptide} or transcription factors switch on {genes / gene for β polypeptide}	(4)